

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** 1,4-Dihydroxyanthraquinone

**CAS No** 81-64-1

**Synonyms** 1,4-Dihydroxy-9,10-anthracenedione; C.I. 58050; Quinizarin

**Product Code** **L04381**

**Address** ThermoFisher Scientific Australia Pty Ltd  
5 Caribbean Drive, Scoresby  
VICTORIA 3179, Australia

**Emergency Tel.** **CHEMTREC®**  
**03 9757 4559 or +613 9757 4559**

**Telephone / Fax Numbers** Tel: 1300 735 292  
Fax: 1800 067 639

**E-mail address** ANZinfo@thermofisher.com

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

No hazards identified

#### Health hazards

No hazards identified

#### Environmental hazards

Acute aquatic toxicity  
Chronic aquatic toxicity

Category 1  
Category 1

#### Label Elements



Environment

**Signal Word****Danger****Hazard Statements**

H410 - Very toxic to aquatic life with long lasting effects

**Precautionary Statements**

P403 - Store in a well-ventilated place

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

## Section 3 - Composition and Information on Ingredients

| Component                          | CAS No  | Weight % |
|------------------------------------|---------|----------|
| 1,4-Dihydroxy-9,10-anthracenedione | 81-64-1 | >95      |

## Section 4 - First Aid Measures

|                                            |                                                                                                                                                  |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                     |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                                            |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                                                           |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                             |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable.                                                                                                                     |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                           |

## Section 5 - Fire Fighting Measures

**Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Hazardous Decomposition Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

**Specific Hazards Arising from the Chemical**

Do not allow run-off from fire-fighting to enter drains or water courses.

**Special protective equipment and precautions for fire fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

**Emergency procedures**

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.

**Environmental Precautions**

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. See Section 12 for additional Ecological Information. Avoid release to the environment. Collect spillage.

**Methods for Containment and Clean Up****Clean-up methods - small spillage**

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

**Clean-up methods - large spillage**

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation.

**Conditions for Safe Storage, Including any Incompatibilities**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

**Exposure limits**

The product does not contain any hazardous materials with occupational exposure limits established.

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-------------------|-----------------|-----------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | AS/NZS 2161     | (minimum requirement) |
| Neoprene       | recommendations   |                 |                 |                       |
| Natural rubber |                   |                 |                 |                       |
| PVC            |                   |                 |                 |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Long sleeved clothing

**Respiratory Protection**

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:**

Particulates filter conforming to EN 143 (or AUS/NZ equivalent)

**Recommended half mask:-**

Particle filtering: EN149:2001 (or AUS/NZ equivalent)

When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## Section 9 - Physical and Chemical Properties

**Information on basic physical and chemical properties**

|                            |                                                            |
|----------------------------|------------------------------------------------------------|
| <b>Appearance</b>          | Red                                                        |
| <b>Physical State</b>      | Powder Solid                                               |
| <b>Odor</b>                | Odorless                                                   |
| <b>Odor Threshold</b>      | No data available                                          |
| <b>pH</b>                  | No information available                                   |
| <b>Melting Point/Range</b> | 191 - 193 °C / 375.8 - 379.4 °F                            |
| <b>Softening Point</b>     | No data available                                          |
| <b>Boiling Point/Range</b> | 450 °C / 842 °F @ 760 mmHg                                 |
| <b>Flash Point</b>         | 222 °C / 431.6 °F <b>Method -</b> No information available |

|                                         |                          |       |
|-----------------------------------------|--------------------------|-------|
| Evaporation Rate                        | Not applicable           | Solid |
| Flammability (solid,gas)                | No information available |       |
| Explosion Limits                        | No data available        |       |
| Vapor Pressure                          | 1 mmHg @ 197 °C          |       |
| Vapor Density                           | Not applicable           | Solid |
| Specific Gravity / Density              | No data available        |       |
| Bulk Density                            | No data available        |       |
| Water Solubility                        | <1 g/L (20°C)            |       |
| Solubility in other solvents            | No information available |       |
| Partition Coefficient (n-octanol/water) |                          |       |
| Component                               | log Pow                  |       |
| 1,4-Dihydroxy-9,10-anthracenedione      | 4.16                     |       |
| Autoignition Temperature                | 315 °C / 599 °F          |       |
| Decomposition Temperature               | No data available        |       |
| Viscosity                               | Not applicable           | Solid |
| Explosive Properties                    | No information available |       |
| Oxidizing Properties                    | No information available |       |
| <b>Other information</b>                |                          |       |
| Molecular Formula                       | C14 H8 O4                |       |
| Molecular Weight                        | 240.21                   |       |

## Section 10 - Stability and Reactivity

|                                  |                                                          |
|----------------------------------|----------------------------------------------------------|
| Reactivity                       | None known, based on information available               |
| Stability                        | Stable under normal conditions.                          |
| Conditions to Avoid              | Incompatible products.                                   |
| Incompatible Materials           | Strong oxidizing agents, Strong acids.                   |
| Hazardous Decomposition Products | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). |
| Hazardous Polymerization         | No information available.                                |

## Section 11 - Toxicological Information

### Information on Toxicological Effects

#### Product Information

|                     |                                                                  |
|---------------------|------------------------------------------------------------------|
| (a) acute toxicity; |                                                                  |
| Oral                | Based on available data, the classification criteria are not met |
| Dermal              | No data available                                                |
| Inhalation          | No data available                                                |

| Component                          | LD50 Oral             | LD50 Dermal               | LC50 Inhalation           |
|------------------------------------|-----------------------|---------------------------|---------------------------|
| 1,4-Dihydroxy-9,10-anthracenedione | LD50 > 5 g/kg ( Rat ) | LD50 > 2500 mg/kg ( Rat ) | LC50 > 1 mg/L ( Rat ) 4 h |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;  
Respiratory No data available

|                                                   |                                                                                |
|---------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Skin</b>                                       | No data available                                                              |
| <b>(e) germ cell mutagenicity;</b>                | No data available                                                              |
| <b>(f) carcinogenicity;</b>                       | No data available<br>There are no known carcinogenic chemicals in this product |
| <b>(g) reproductive toxicity;</b>                 | No data available                                                              |
| <b>(h) STOT-single exposure;</b>                  | No data available                                                              |
| <b>(i) STOT-repeated exposure;</b>                | No data available                                                              |
| <b>Target Organs</b>                              | No information available.                                                      |
| <b>(j) aspiration hazard;</b>                     | Not applicable<br>Solid                                                        |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated.                 |
| <b>Symptoms / effects, both acute and delayed</b> | No information available                                                       |

## Section 12 - Ecological Information

|                                       |                                                                                                                                                                                     |            |                                   |          |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------------------------------|----------|
| Ecotoxicity effects                   | Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment. |            |                                   |          |
| Component                             | Freshwater Fish                                                                                                                                                                     | Water Flea | Freshwater Algae                  | Microtox |
| 1,4-Dihydroxy-9,10-anthracenedione    |                                                                                                                                                                                     |            | EC50: 0,44 mg/L/72h (green algae) |          |
| Persistence and Degradability         | Not readily biodegradable                                                                                                                                                           |            |                                   |          |
| Degradation in sewage treatment plant | Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.                                                                     |            |                                   |          |
| Bioaccumulative Potential             | No information available                                                                                                                                                            |            |                                   |          |
| Component                             | log Pow                                                                                                                                                                             |            | Bioconcentration factor (BCF)     |          |
| 1,4-Dihydroxy-9,10-anthracenedione    | 4.16                                                                                                                                                                                |            | No data available                 |          |
| Mobility                              | No information available.                                                                                                                                                           |            |                                   |          |
| Endocrine Disruptor Information       | This product does not contain any known or suspected endocrine disruptors                                                                                                           |            |                                   |          |
| Persistent Organic Pollutant          | This product does not contain any known or suspected substance                                                                                                                      |            |                                   |          |
| Ozone Depletion Potential             | This product does not contain any known or suspected substance                                                                                                                      |            |                                   |          |

## Section 13 - Disposal Considerations

|                                            |                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations. |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point.                                                                                                                                                                                                                                            |
| <b>Other Information</b>                   | Chemical wastes should be disposed through a licensed commercial waste collection service. Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.                      |

## Section 14 - Transport Information

### IMDG/IMO

UN-No UN3077  
Proper Shipping Name Environmentally hazardous substances, solid, n.o.s.  
Technical Shipping Name 1,4-Dihydroxy-9,10-anthracenedione  
Hazard Class 9  
Packing Group III

### ADG

UN-No UN3077  
Proper Shipping Name Environmentally hazardous substances, solid, n.o.s.  
Technical Shipping Name 1,4-Dihydroxy-9,10-anthracenedione  
Hazard Class 9  
Packing Group III

### IATA

UN-No UN3077  
Proper Shipping Name Environmentally hazardous substances, solid, n.o.s.  
Technical Shipping Name 1,4-Dihydroxy-9,10-anthracenedione  
Hazard Class 9  
Packing Group III

Environmental hazards Dangerous for the environment  
Product is a marine pollutant according to the criteria set by IMDG/IMO

Special Precautions No special precautions required

Additional information None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations Australia

See section 8 for national exposure control parameters.

#### **Standard for the Uniform Scheduling of Medicines and Poisons**

No poison schedule number allocated.

#### **Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component                                    | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|----------------------------------------------|-------------------------------------------------------------|------------------------|
| 1,4-Dihydroxy-9,10-anthracenedione - 81-64-1 | Present                                                     | -                      |

#### **Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

**Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

**National pollutant inventory** Not applicable

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

**International Inventories**

| Component                          | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|------------------------------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| 1,4-Dihydroxy-9,10-anthracenedione | X    | X     | 201-368-7 | -      | X    | X   | -    | X     | X    | X    | X     | KE-10777 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Basel convention on the control of transboundary movements of hazardous wastes and their disposal**

Not applicable.

| Component                          | CAS No  | OECD HPV | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|------------------------------------|---------|----------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 1,4-Dihydroxy-9,10-anthracenedione | 81-64-1 | Listed   | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

**Authorisation/Restrictions according to EU REACH** Not applicable

**Section 16 - Other Information****Legend**

**AICS** - Australian Inventory of Chemical Substances  
**TSCA** - United States Toxic Substances Control Act Section 8(b)

**NZIoC** - New Zealand Inventory of Chemicals

## Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**NZS 5433:2012** - Transport of Dangerous Goods on Land

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**WEL** - Workplace Exposure Limit

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists  
Predicted No Effect Concentration (PNEC)

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**ADG** Australian Code for the Transport of Dangerous Goods by Road and Rail

**OECD** - Organisation for Economic Co-operation and Development

**LC50** - Lethal Concentration 50%

**ATE** - Acute Toxicity Estimate

**RPE** - Respiratory Protective Equipment

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

**Revision Date**

18-Nov-2022

**Revision Summary**

Not applicable.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**